How Researchers Claim Novelty in Biomedical Science: A Taxonomy for Understanding Innovation

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Introduction

Scientific novelty drives progress, particularly in biomedicine where new insights can transform healthcare. Despite its importance, novelty remains poorly defined and inconsistently categorized. Existing definitions often emphasize limited dimensions (e.g., temporal or conceptual novelty)[1,2], lacking a comprehensive schema to assess biomedical contributions systematically. To address this, we introduce *NovelTax*, a structured, hierarchical taxonomy capturing diverse manifestations of novelty across the biomedical literature.

Methods and Materials

We developed *NovelTax* through a data-driven, iterative, and LLM-assisted process involving three phases (Fig. 1):

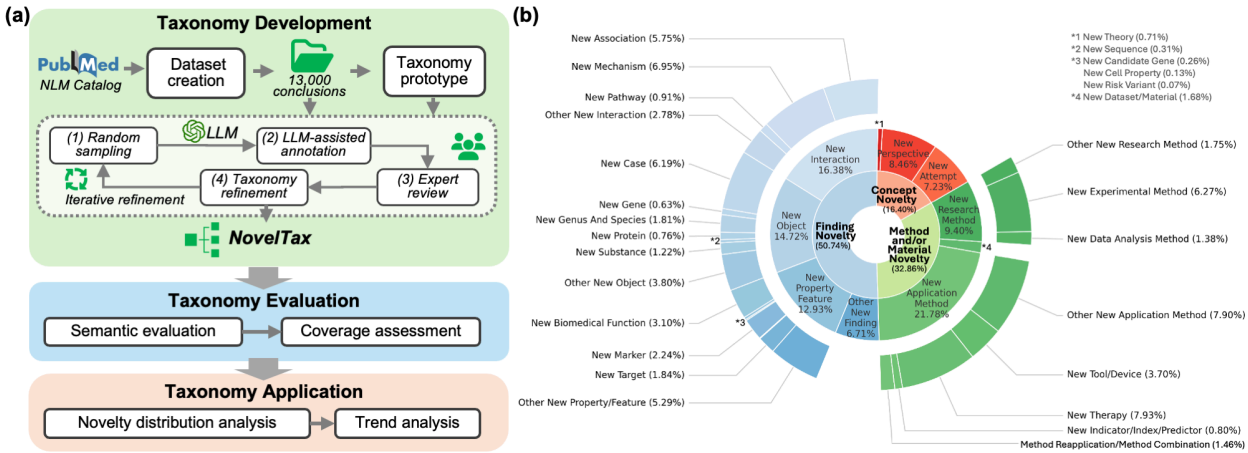

Fig. 1. (a) The overall workflow of the taxonomy construction, including taxonomy development, taxonomy evaluation, and taxonomy application. **(b)** Novelty distribution on *Biomedical Novelty Claim Corpus*.

Taxonomy Development. We curated a *Biomedical Novelty Claim Corpus* of 12,701 PubMed conclusions containing self-reported novelty claims (published between 1781 and 2024). These spanned 130 subdomains, sampled via Broad Subject Terms. An initial taxonomy was designed through expert discussions and refined through repeated GPT-4o labeling and human adjudication cycles, reaching an annotation accuracy of 0.84.

Taxonomy Evaluation. We validated *NovelTax* both semantically (via semantically equivalent natural language questions and expert consensus; rational score = 0.81, agreement = 0.92) and through coverage on unseen samples (88% coverage, agreement = 0.86).

Taxonomy Application.We applied *NovelTax* to analyze novelty trends across biomedical subdomains, using the full corpus to reveal dominant innovation types.

Results

NovelTax comprises 39 nodes across 3 levels, organized into Concept novelty, Method and/or material novelty, and Finding novelty. Most biomedical articles (50.74%) fall under Finding novelty, reflecting the field's strong focus on empirical discoveries (e.g., new mechanisms or associations). In contrast, Concept novelty (e.g., new theories) remains rare.

Conclusion and Future Work

NovelTax offers a validated, fine-grained framework for understanding and categorizing biomedical novelty. Its broad coverage, hierarchical depth, and practical application make it a valuable tool for identifying emerging trends, informing strategic funding, and guiding scientific exploration. Future work includes extending *NovelTax* to subdomain-specific trend analyses and integrating it with other knowledge graphs and ontologies.

References

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